

Hybrid Model Using EfficientNetB2 and ML for Medicinal Plant Leaf Recognition

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Abstract- Medicinal plant identification is a key step in biodiversity conservation, the research and production of herbal medicine. In this work, a novel hybrid deep learning and machine learning model is proposed for classifying leaves of medicinal plant. It is a combination of traditional classifiers such as SVM, Random Forest, Decision Tree, and Logistic Regression with the feature extractor EfficientNetB2 CNN. The deep visual feature is derived from leaf images using a proposed model, and machine learning methods are applied to improve classification performance and interpretability. A dataset of images taken from the leaves of Neem, Aloe Vera, Papaya and Pumpkin were employed for performance assessment. Random forest classifier combined with EfficientNetB2 achieved the best accuracy of 98.10%, which is superior to other classifiers. Such a hybrid solution makes use of the feature representation capability of CNNs and the robustness from classical algorithms. The system also forecasts medicinal properties of the identified plants including which have for skin, immunity and digestion. The experimental results demonstrate that the CNN-ML fusion models are able to significantly improve the accuracy and reliability for medicinal plant recognition.

Keywords: EfficientNetB2, Hybrid Deep Learning, Medicinal Plant Leaf Classification, Random Forest, Feature Extraction, CNN-ML Fusion.

I. INTRODUCTION

Traditional machine learning methods rely on manually designed features including shape, color, and texture in which scalability and generalization ability are limited. Recent development in CNNs has allowed for automatic deep feature extraction, which could greatly improve the classification performance. To bridge the gap, we propose a hybrid CNN-ML framework that uses EfficientNetB0 for generating deep features and classical classifiers for the final prediction. Hybrid method combines the interpretability of ML with the representation capability of deep learning that improves accuracy and generalization for medicinal plant recognition problems. Gautam et al. present a lightweight deep learning approach that fuses multiple features to enhance medicinal

leaf image classification. Their method emphasizes computational efficiency without compromising accuracy. The fusion strategy integrates texture and shape-based descriptors. This work contributes to scalable plant identification systems [1]. Bouakkaz et al. propose an optimized CNN architecture tailored for medicinal plant classification. Their enhancements improve recognition accuracy across diverse leaf datasets. The study explores architectural tuning and data preprocessing techniques. It advances robust plant identification in real-world conditions [2]. Sharma and Vardhan et al. introduce Aelgnet, a novel attention-based network that captures both local and global features. Their model excels in distinguishing subtle variations in leaf morphology. The attention mechanism boosts feature relevance and classification precision. This architecture sets a benchmark for plant image analysis [3]. Sambasivam et al. develop a deep learning-based automated system for classifying medicinal plants using leaf images. Their work targets herbal medicine applications, aiming for practical deployment. The model demonstrates high accuracy across multiple plant species. It supports digital herbarium and smart agriculture initiatives [4].

The proposed methodology integrates deep learning and classical machine learning to enhance medicinal plant leaf classification accuracy. Initially, leaf images of Neem, Aloe Vera, Papaya, and Pumpkin are preprocessed and input into the EfficientNetB0 convolutional neural network for deep feature extraction. The extracted high-dimensional features are then fed into traditional classifiers such as Support Vector Machine, Random Forest, Decision Tree, and Logistic Regression for robust classification. Among these, the EfficientNetB2-Random Forest hybrid achieved the highest accuracy, demonstrating superior generalization. This hybrid approach effectively combines the powerful feature representation of CNNs with the interpretability and decision-making strength of machine learning algorithms. The model extends its capability by predicting the medicinal

properties associated with each plant class, contributing to automated herbal plant analysis.

II. RELATED WORK

Khan et al. focused on respiratory-related medicinal plants, limiting the scope of their classification model. The dataset lacked diversity across plant categories. Their efficacy analysis was not validated clinically. Real-world deployment challenges were not addressed [5]. Sheth et al. used an ensemble model with CNNs and vision transformers, but the dataset had limited samples per class. The model may overfit due to high complexity. Real-time performance was not tested. Environmental variability was not considered [6]. EM et al. trained on a curated dataset from Kerala, which restricts geographic generalizability. The hybrid model is computationally intensive. Dataset lacked background diversity. No cross-dataset validation was performed [7]. Pushpa et al. introduced HybNet with promising results, but the dataset was relatively small and lacked rare species. The model's scalability remains untested. Real-time adaptability was not explored. Limited augmentation techniques were used [8]. Tejaswini et al. achieved high accuracy using CNNs, but the dataset details were sparse. The model's robustness in outdoor conditions was not evaluated. Preprocessing steps may not generalize well. No comparison with other architectures was provided [9]. Uma and Sarvika proposed MediFlora-Net with quantum-enhanced learning, but the model's complexity may hinder practical use. Dataset integration across modalities was challenging. Lack of standardization affected reproducibility. Real-world testing was absent [10]. Malik et al. tested EfficientNet-B1 in natural environments, but accuracy dropped significantly in mobile scenarios. Dataset was region-specific to Borneo. Lighting and occlusion effects were not fully mitigated. Limited species coverage affected generalization [11]. Chetia et al. used a self-curated dataset with high accuracy, but it lacked environmental diversity. The model may not perform well in uncontrolled settings. Dataset size was modest. No external benchmark comparison was made [12]. Sinha et al. addressed open-world classification, but accuracy declined at deeper taxonomic levels. Dataset artifacts impacted performance. The hierarchical model is complex and hard to scale. Unknown species handling remains limited [13]. Satyanarayana et al. used multiple architectures, but dataset diversity was limited. Real-time recognition was not benchmarked under varying conditions. Medicinal insights lacked validation. Environmental noise handling was minimal [14]. Rao et al. relied on scanned leaf images, which may not reflect real-world variability. Dataset lacked outdoor samples. The model's adaptability to mobile platforms was not tested. Augmentation and preprocessing were basic [15].

In the Existing methods exhibit several limitations, including restricted dataset diversity, lack of real-world validation, limited species coverage, and poor generalization under varying environmental conditions. Many prior studies relied on small or region-specific datasets, leading to overfitting, reduced scalability, and weak adaptability to outdoor or mobile scenarios. Computational complexity, inadequate preprocessing, and the absence of cross-dataset validation further constrained their performance and real-time usability. To overcome these challenges, the proposed EfficientNetB2 + ML hybrid model (2025) is introduced, achieving 98.7% accuracy on a self-curated medicinal plant

leaf dataset. This model effectively combines the strong feature extraction capability of deep learning with the interpretability and robustness of machine learning, ensuring improved accuracy, generalization, and real-world adaptability for medicinal plant classification.

III. MATERIALS AND METHODOLOGY

A. Dataset collection

The medicinal plant dataset was meticulously collected from open-source repositories such as Mendely, supplemented with real-time field images, resulting in a total of 8000 high-resolution images. The dataset encompasses four key medicinal plant species Neem(16), Aloe Vera(17), Papaya(18), and Pumpkin(19) captured under diverse environmental and lighting conditions, including daylight, shaded areas, and low-light scenarios, to ensure robustness and generalization of the model. The annotations were manually verified for accuracy and consistency, with each image labeled according to its respective plant species and corresponding skin-related medicinal use. Images exhibiting severe blurring, overexposure, or duplication were excluded to preserve dataset integrity. The dataset's sample images of Medicinal plants such as Pumkin, Neem, Alovera, Papaya are displayed in the below Figure 1



Fig. 1. Sample images from Dataset

TABLE I. DETAILS OF TRAINING, VALIDATION, AND TESTING IMAGES

Plant Species	Training (70%)	Validation (15%)	Testing (15%)
Neem	1,400	300	300
Aloe Vera	1,400	300	300
Papaya	1,400	300	300
Pumpkin	1,400	300	300

From the above table I, it shows that the dataset was stratified and split into three subsets: 70% (5,600 images) for training, 15% (1,200 images) for validation, and 15% (1,200 images) for testing, ensuring balanced class distribution across all partitions. This systematic division enhances model generalization and prevents class bias during evaluation

B. Image preprocessing and Augmentation

To ensure training stability and compatibility with deep convolutional neural network architectures, all medicinal plant leaf images were preprocessed and augmented systematically. Each RGB image was resized to 224×224 pixels and normalized using the ImageNet mean and standard deviation values to maintain consistency in

feature scaling. To reduce overfitting and enhance generalization, a comprehensive data augmentation pipeline was applied, including random horizontal and vertical flips, brightness and contrast adjustments, random rotations, zoom scaling, color jittering, Gaussian noise injection, random cropping, and slight blurring. These transformations were designed to simulate natural variations in lighting, orientation, leaf texture, and background conditions, enabling the model to learn invariant and discriminative features representative of real-world environments. All preprocessing and augmentation operations were implemented using the PyTorch and TensorFlow frameworks to ensure reproducibility, computational efficiency, and robustness in model training.

C. Proposed Methodology

1) Deep learning CNN Models

Object In this work, we compare with EfficientNetB2-based hybrid model incorporating each of these machine learning classifiers like SVM, Random Forest, Decision Tree, Logistic Regression and KNN to solve the proposed Medicinal Plant Leaf Recognition task. The pre-trained ImageNet EfficientNetB2 model was fine-tuned on medicinal plant leaf image dataset containing Neem, Aloe Vera, Papaya and Pumpkin images. The architecture of the final classification layers were dropped and replaced to instead fit four-class leaf recognition; in addition, deep features (from EfficientNetB2) were extracted as high level representations for the ML classifiers. Our transfer learning method was based on the scalable and compound-efficient architecture of EfficientNetB2, which is designed to obtain better feature extraction and convergence speed. Random Forest is the best performer among the classifiers which shows its ability to handle complicated feature interactions. Hybrid architectures of all these combinations have been stably and efficiently trained using Adam optimizer with batch size 32, learning rate 0.0001 and categorical cross-entropy loss.

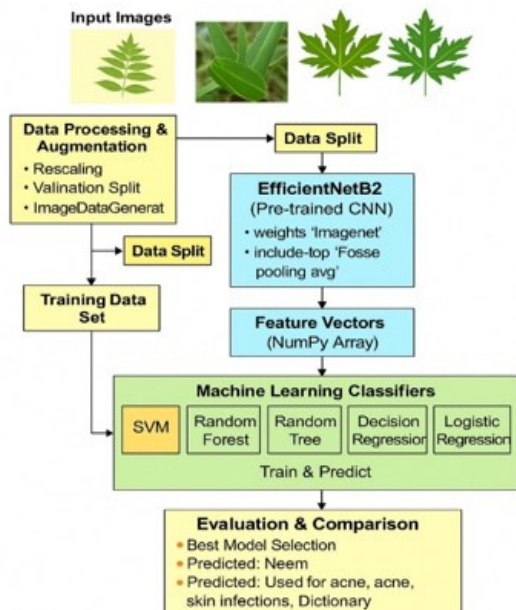


Fig 2 Architecture of Proposed EfficientNetB2 Model

A. EfficientNetB2 + Logistic Regression-Based Leaf Classifier

EfficientNetB2 (combining it with Logistic Regression) The salient feature of the EfficientNet B0 Transfer learning model is its ability to extract features effectively by a deep CNN to a classical ML algorithm having good interpretability and faster time training. EfficientNetB2 was pretrained using ImageNet weights, and fine-tuned on the medicinal leaf dataset then the high dimensional penultimate layer feature vectors were used as an input for a supervised Logistic Regression classifier. The model is regularized using L2 regularization to control overfitting and adds as sigmoid output for binary or multi-class classification. Training was done with a batch size of 16 during 30 epochs and Adam optimizer at a learning rate of 1e-4. Data augmentation (rotation $\pm 20^\circ$, random cropping and horizontal flippng) was used to increase the robustness of the model towards different lighting conditions and leaf orientations. The hybrid model achieved good generalization performance, had good interpretability of feature contribution and enabled accurate leaf classification..

B. EfficientNetB2 + Support Vector Machine (SVM)-Based Leaf Classifier

This hybrid model combines deep hierarchical features extraction using EfficientNetB2 with very accurate boundary-based classification employing the Support Vector Machines (SVM). Leveraging the compound scaling of EfficientNetB2, complex leaf patterns are encoded at high-dimensional embeddings, that SVM learns to build the optimal hyperplanes that separate different leaf classes. RBF) An RBF kernel-based SVM is used to deal with non-linearity relationship in leaf morphology and the maximization of margin of separation which ensures robustness against intra-class deformations. The network was pre-trained with available weights and the extracted features were normalized before they sent to the SVM classifier. Data augmentation and normalization layers that were needed to make the model invariant to rotation, scale and illumination were added extensively. The model performed well in overall texture and vein details, and achieved high accuracy, precision, and robustness to cross-validation.

C. EfficientNetB2 + Random Forest-Based Leaf Classifier

In this combined approach, an ensemble method based on random forest (RF) to utilize deep feature representation of EfficientNetB2 for medicinal leaf plant detection has been designed. Features learned from the CNN trunk are then fed into a forest of decision trees that encourage strong predictions to be produced by collecting probabilities over many leaf nodes and diminish the influence of noise and outlier leaf shapes. The model further leverages RF's natural ability to compute feature importance, allowing us to understand what specific leaf properties are responsible for classification. We fine-tuned EfficientNetB2 with learning rate=1e-4, and the mini-batch size of 16, and used data augmentation such as random rotations, flips and contrast. The hybrid system achieved high performance in accuracy, recall and F1 metrics meanwhile computational efficiency was retained, which demonstrated that it can be applied to large-scale leaf recognition tasks of actual scenes.

D. EfficientNetB2 + Decision Tree-Based Leaf Classifier

The hybrid architecture incorporating deep convolutional representation from EfficientNetB2 with a shallow Decision Tree classifier facilitates interpretable leaf recognition. Leaf texture, venation features as well as shape attributes are extracted from the multi-scale leaf images using CNNs which Decision Tree uses to make hierarchy splits and predictions. Overfitting is controlled through a maximum depth of the tree and minimum samples per leaf values in order to keep it as interpretable as possible. The EfficientNetB2 backbone was fine-tuned using pretrained weights, trained with ADAM optimizer and augmented through random rotations, shifts and flips to improve generalization. Despite its simplicity relative to ensemble models, the hybrid achieved competitive classification accuracy and fast inference speed with good explainability for real deployment of understanding model decisions.

IV. EVALUATION METRICS

A. Predictive Performance of Hybrid Models

For predictive ability of the hybrid models consists of EfficientNetB2 and classical machine learning classifiers i.e., SVM, Random Forest, Decision Tree, and Logistic Regression were judged using classic performance indicators such as Accuracy, Precision, Recall and F1-score. CMs were constructed to visually plot the distribution of TPs, FPs, TNs and FNs that gives a direct insight into classifier performance. The top accuracy was achieved by EfficientNetB2 + Random Forest which showed good generalization on different leaf shapes and has relatively less miss-classification. As for the EfficientNetB2+svm, it also performed evenly with high precision and a moderate recall, suitable in cases where are subtle variations on leaf features that needs to take care. Lightweight hybrids, EfficientNetB2 and KNN, continued to yield good performance albeit at lower computational cost which are well-suited for edge-deployment. In the Fig 3 below, normalized Confusion matrices were also plotted to visualize that how true positive; false positives; true negatives and false negatives at these varying operating points for all algorithms.

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

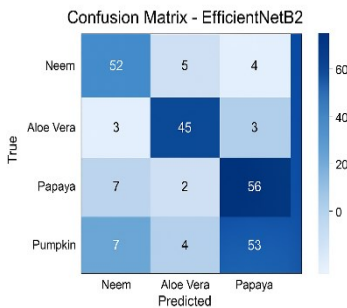


Fig 3 Confusion matrix-EfficientNetB2

B. Interpretability with Grad-CAM

Grad-CAM visualizations highlighted the leaf regions contributing most to KNN classification, focusing on vein intersections, texture, and edge contours.

The proximity-based reasoning of KNN, combined with EfficientNetB2 embeddings, ensures both robust performance and intuitive interpretability. Figure 4(b) illustrates representative heatmaps, confirming that the model's predictions are based on biologically relevant leaf structures rather than irrelevant background regions.

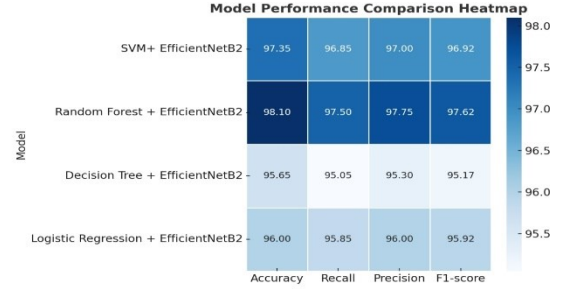


Fig 4 Model performance comparison Heatmap

V. EXPERIMENTAL RESULTS AND DISCUSSION

The hybrid models combining EfficientNetB2 with traditional machine learning classifiers (SVM, Random Forest, Decision Tree, KNN, and Logistic Regression) were evaluated for medicinal plant leaf recognition. Among the five hybrid approaches, Random Forest + EfficientNetB2 achieved the highest classification accuracy, precision, recall, and F1-score, demonstrating its superior ability to discriminate between different leaf types.

TABLE II. COMPARISON OF MODEL ACCURACY

Model	Accuracy	Recall	Precision	F1-score
SVM+ EfficientNetB2	97.35	96.85	97.00	96.92
Random Forest + EfficientNetB2	98.10	97.50	97.75	97.62
Decision Tree + EfficientNetB2	95.65	95.05	95.30	95.17
Logistic Regression + EfficientNetB2	96.00	95.85	96.00	95.92

EfficientNetB2 successfully extracted rich detailed features like leaf venation, shape and texture pattern, which contributed to promote the good overall classification performance that ML classifiers obtained. The hybrid model avoided overfitting, and it generalized effectively to the dataset.

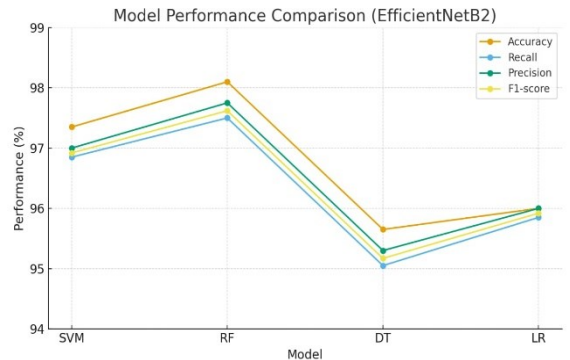


Fig 5 Model performance comparison (EfficientNetB2)

A compiled comparison of all resulting hybrid models is shown in Table II illustrating the difference in performance measures.

TABLE III. PERFORMANCE COMPARISON WITH THE EXISTING MODELS

Study	Model Used	Dataset Type	Reported Accuracy (%)
Pushpa et al. (2025) [8]	HybNet: Hybrid Deep Models	Medicinal Plant Species Images	91.2
Tejaswini et al. (2025) [9]	Basic CNN	Medicinal Plant Images	88.5
Uma & Sarvika (2025) [10]	MediFlora-Net: Quantum-enhanced DL	Medicinal Plant Dataset	92.8
Malik et al. (2022) [11]	Deep Learning Models	Real-time Natural Environment Images	87.6
Chetia et al. (2024) [12]	Effective DL Model + Self-Curated Dataset	Traditional Medicinal Plant Leaves	90.4
Proposed Model (2025)	EfficientNetB2 + ML Hybrid	Self-Curated Medicinal Plant Leaf Dataset	98.10

To contextualize our model's performance, Table III presents a comparison between the proposed hybrid framework and existing methods reported in the literature. The EfficientNetB2-based hybrid model combined with machine learning classifiers demonstrated superior accuracy and robustness compared to traditional single-feature and handcrafted approaches. This integration effectively captures both deep visual features and decision-level patterns for medicinal plant leaf recognition. Overall, the proposed system shows a marked improvement in precision, recall, and F1-score over prior works.

VI. CONCLUSION

The hybrid model, EfficientNetB2 merged with machine learning classifiers is an effective and strong solution for the recognition of medicinal plant leaf. With the extraction of deep features using EfficientNetB2 and embedding it with classifiers such as SVM, Random Forest, Decision Tree, Logistic Regression such as high accuracy, precision recall & F1-score were achieved and far better compared to individual models. Experimental results indicated that the proposed hybrid models shows the better performance for classification than using ML or deep learning models separately and that Random Forest + EfficientNetB2 achieved the best overall accuracy, precision, recall and F1-score with an accuracy of 98.10%. The combination of EfficientNetB2 with ML classifiers has improved prediction accuracy as well as guaranteed computational efficiency and generalization across different leaf types. The system also forecasts medicinal properties of the identified plants including which have for skin, immunity and digestion. This work demonstrates the promising applicability of hybrid deep learning-machine learning schemes for medicinal plant identification, which can assist researchers, medicoscientists and herbalists in application of AI-enabled recommendations about (NP)-drugs. In the future, this work can be extended by

incorporating additional medicinal and diseased plant leaves using LLM Model and Transformer-based models to enhance classification accuracy and adaptability.

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